Ice Chamber Experiment Metagenome

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Abstract

Sea ice and its associated frost flowers host taxonomically and functionally distinct microbial communities that play important roles in polar biogeochemical cycles, yet field-based studies of these systems are complicated by confounding regional geography and large-scale ocean and atmospheric circulation, which obscure the local drivers of microbial colonization (Priest et al., 2023). To address this limitation, we used the Roland von Glasow Air-Sea-Ice Chamber, a large laboratory-scale sea ice mesocosm at the University of Grenoble Alpes, to characterize bacterial and eukaryotic community assembly across frost flower, brine, ice matrix, and seawater niches under controlled conditions. A total of 124 SRA runs from 45 samples were processed using an amplicon sequence variant (ASV)-based bioinformatics pipeline, including quality filtering, taxonomic classification, and functional annotation. Of 84 samples subjected to ASV inference, an initial 860,222 reads were quality filtered to 519,769 retained reads (60.4% retention). Analysis of the resulting community profiles revealed taxonomic and functional partitioning of bacteria into distinct niches consistent with niche-based selection, supported by the emergence of archetypal sea-ice bacterial taxa that were not detected in the seeding seawater. In contrast, eukaryotic community composition remained largely consistent across the ice-brine-seawater profile, suggesting that eukaryotic distribution in this system was governed more by stochastic processes than by deterministic niche selection. Bacterial enrichment within the ice matrix proceeded at a rate comparable to that reported in prior field-based observations, despite minimal diatom representation in this experimental system. These findings demonstrate that controlled mesocosm experiments can effectively isolate local physicochemical drivers of microbial colonization from confounding environmental variability, and support a model in which bacterial and eukaryotic communities in sea ice are shaped by distinct assembly processes—niche-based selection and ecological stochasticity, respectively.

Introduction

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Sea ice and its associated frost flowers—delicate, salt-rich structures that form on the surface of newly frozen sea ice—host diverse microbial communities that contribute meaningfully to polar biogeochemical cycles, including carbon flux, nutrient recycling, and the production of climate-active aerosols [ref15]. Despite their ecological importance, these communities are challenging to study in natural settings, where large-scale atmospheric and oceanic dynamics confound efforts to isolate the local, deterministic forces governing microbial colonization and assembly [ref14]. Distinguishing niche-based selection from stochastic dispersal processes is a central question in microbial ecology [ref13], yet field studies of sea ice are inherently limited in their ability to control for the many interacting physical and chemical gradients that develop during ice formation. Laboratory-scale simulations offer an alternative: the Roland von Glasow Air-Sea-Ice Chamber, a large-scale experimental facility capable of generating young sea ice and frost flowers under controlled conditions, provides an opportunity to examine microbial colonization while minimizing the confounding influence of regional atmospheric and oceanic variability.

In this study, we used the Air-Sea-Ice Chamber at the University of Grenoble-Alpes to investigate how bacterial and eukaryotic assemblages are structured across the distinct physicochemical niches that develop within experimentally generated sea ice—frost flowers, brine, the ice matrix, and the underlying seawater. Building on prior observations that bacterial taxa in sea ice environments often display strong niche-specific enrichment [ref12], we hypothesized that bacterial communities in the chamber system would exhibit significant taxonomic partitioning among these habitats, consistent with deterministic niche-based selection. We further predicted that this partitioning would be evidenced by the emergence of archetypal sea-ice bacterial taxa that were not present in the seawater used to seed the experiment. By contrast, we anticipated that eukaryotic assemblages would show comparatively little variation across the ice profile, reflecting a greater role for stochastic colonization processes among these taxa. Finally, we expected that enrichment of characteristic sea-ice bacteria would be detectable even under conditions of minimal diatom representation, which would underscore the robustness of bacterial responses to ice-associated physicochemical conditions independent of eukaryotic primary production.

To test these hypotheses, we reanalyzed publicly available 16S rRNA gene amplicon sequencing data generated from the Air-Sea-Ice Chamber experiment (BioProject PRJNA1473294), comprising 45 biological samples and 124 sequencing runs spanning the frost flower, brine, ice matrix, and seawater niches. Raw sequence reads were processed using the microscape pipeline, which incorporates quality filtering, primer removal, denoising via DADA2, chimera removal, taxonomic classification against the SILVA reference database, and decontamination of likely contaminant sequences. The resulting amplicon sequence variant table was used to characterize taxonomic composition across niches and to evaluate patterns of alpha- and beta-diversity, differential taxon abundance, and co-occurrence structure. This approach allowed us to assess, at high resolution, the extent to which bacterial and eukaryotic community structure in this experimentally controlled sea-ice system reflects deterministic niche selection versus stochastic assembly processes.

Methods

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Data Acquisition

Sequence data analyzed in this study were derived from an experimental sea ice mesocosm study conducted using the Roland von Glasow Air-Sea-Ice Chamber (University of Grenoble Alpes), designed to isolate local microbial colonization dynamics on frost flowers, brine, ice matrix, and seawater from confounding regional environmental effects. A total of 45 samples were processed, corresponding to 124 sequencing runs deposited under BioProject PRJNA1473294 in the NCBI Sequence Read Archive (SRA). Samples represented air metagenome and three additional sample types/environments consistent with the frost flower, brine, ice matrix, and seawater niches described in the study design [AUTHOR: confirm sample-to-environment mapping and per-environment sample counts]. Raw sequencing reads were retrieved from SRA using standard SRA toolkit utilities prior to downstream processing.

Sequence Processing Pipeline

Amplicon sequence data were processed using the microscape pipeline, which implements an amplicon sequence variant (ASV)-based workflow analogous to DADA2-style denoising. Following quality filtering, an initial set of 84 samples with sufficient read depth were retained for downstream community analysis (out of the total 45 biological samples/124 sequencing runs, indicating that multiple runs were merged or samples were processed at the run level for this step [AUTHOR: clarify sample-to-run aggregation strategy]). Quality filtering removed low-quality and chimeric reads, reducing the initial 860,222 input reads to 519,769 retained reads, corresponding to an overall retention rate of 60.4%.

Filtered reads were used to construct sequence tables (seqtabs) and, following merging and chimera removal, a final ASV table (seqtab_final) was generated. Taxonomic classification of ASVs was performed against the SILVA reference database, which is the sole taxonomy database employed in this analysis (version not specified in pipeline metadata) [AUTHOR: specify SILVA release version if known]. Classification was conducted hierarchically across taxonomic ranks from domain to genus.

Analysis Parameters

A total of 162 ASVs were classified using the SILVA database. Classification success was high and consistent across upper taxonomic ranks, with all 162 ASVs (100%) assigned at the domain, phylum, and class levels. Assignment rates declined marginally at finer resolution, with 161 ASVs (99.4%) classified at order and family levels, and 152 ASVs (93.8%) classified at the genus level, reflecting the expected reduction in reference database resolution at lower taxonomic ranks.

At the phylum level, the classified community was dominated by Pseudomonadota (n = 116 ASVs), followed by Bacteroidota (n = 22), Campylobacterota (n = 13), Verrucomicrobiota (n = 4), and Actinomycetota (n = 3). These five phyla accounted for the entirety of the top-phylum classifications reported by the pipeline.

Following taxonomic assignment, ASV abundance tables were renormalized to account for differences in sequencing depth across samples (renormalized output category), enabling comparative analysis of community composition across sample types. Hierarchical clustering was performed on renormalized community data to identify groupings of samples or taxa based on compositional similarity (clustering output category), and co-occurrence network analysis was conducted to characterize potential taxon-taxon associations across the sampled niches (network output category). Specific clustering distance metrics, linkage methods, and network construction thresholds (e.g., correlation cutoffs) were not explicitly reported in the pipeline metadata and are therefore not detailed here [AUTHOR: provide clustering/network parameters if available].

Statistical Approaches

Quality control metrics, including read retention rates and per-sample sequencing depth, were assessed to ensure data adequacy for downstream compositional analyses (quality_check output category). Taxonomic and compositional visualizations (viz output category) were generated to support comparison of bacterial community structure across the four experimental niches (frost flower, brine, ice matrix, and seawater) described in the study. Given the exploratory nature of the pipeline outputs provided, specific statistical tests for differential abundance, alpha/beta diversity, or niche partitioning (e.g., PERMANOVA, ANOSIM) were not explicitly detailed in the available metadata; where such analyses were performed, methodological details should be specified separately

[AUTHOR: provide statistical test details, including diversity metrics and significance thresholds used to support claims of niche-based bacterial partitioning versus stochastic eukaryotic community structure].

All bioinformatic processing was performed using the microscape pipeline; specific tool versions and package dependencies underlying individual pipeline steps (e.g., denoising, chimera removal, clustering algorithms) were not specified in the provided outputs and are noted here as a limitation of the current methods description [AUTHOR: provide software version information for full reproducibility].

Results

Results

Sequence Processing and Quality Filtering

A total of 124 SRA runs derived from 45 samples collected across the Roland von Glasow Air-Sea-Ice Chamber experimental system were processed through the bioinformatics pipeline. Amplicon sequence variant (ASV) inference and quality filtering were applied to 84 samples, yielding an initial dataset of 860,222 reads. Following quality filtering, 519,769 reads were retained, corresponding to a retention rate of 60.4% (Table 1). Filtered sequence tables, clustering outputs, and renormalized abundance tables were generated as part of the downstream processing pipeline (see result categories: filtered, clustering, renormalized, seqtab_final).

Taxonomic Classification

Taxonomic assignment was performed using the SILVA reference database. A total of 162 ASVs were classified at the domain, phylum, and class levels (162/162, 100% at each rank). Classification success declined marginally at finer taxonomic resolution: 161 of 162 ASVs (99.4%) were classified at the order and family levels, and 152 of 162 ASVs (93.8%) were classified at the genus level (Table 2, Figure 1).

Phylum-Level Community Composition

Among the 162 classified ASVs, five bacterial phyla accounted for the majority of taxonomic assignments (Figure 2). Pseudomonadota was the most represented phylum, comprising 116 ASVs (71.6% of classified ASVs), followed by Bacteroidota (22 ASVs, 13.6%), Campylobacterota (13 ASVs, 8.0%), Verrucomicrobiota (4 ASVs, 2.5%), and Actinomycetota (3 ASVs, 1.9%). Together, these five phyla accounted for 158 of the 162 classified ASVs (97.5%), with the remaining ASVs ($n = 4$) distributed among less abundant or unresolved phyla.

Data Availability and Downstream Analyses

Sequence tables, taxonomic classifications, and diversity network outputs are provided in the accompanying supplementary files, organized according to the result categories generated by the pipeline (seqtabs, taxonomy, network, viz, quality_check). These outputs form the basis for the compositional comparisons across sample types (frost flower, brine, ice matrix, and seawater) presented in the following sections.

[AUTHOR: Please provide additional quantitative results specific to niche-based partitioning across frost flower, brine, ice matrix, and seawater sample types, including any statistical comparisons (e.g., diversity indices, differential abundance testing, or ordination results such as PCoA/NMDS) to complete this section. The current pipeline output summary does not include sample-type-resolved taxonomic or diversity comparisons, eukaryotic community composition data, or functional/metagenomic annotation results referenced in the study abstract.]

Discussion

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Summary of Main Findings

This study applied amplicon-based metagenomic sequencing to samples collected from the Roland von Glasow Air-Sea-Ice Chamber, a laboratory-scale experimental system designed to isolate the local physicochemical drivers of microbial colonization in sea ice environments from confounding factors associated with field-based studies, such as regional geography and large-scale ocean and atmospheric circulation [ref11]. From 124 SRA runs across 45 samples, quality filtering and ASV inference yielded 519,769 reads (60.4% retention) from an initial pool of 860,222 reads, and taxonomic classification using the SILVA reference database resolved 162 ASVs with high confidence at broad taxonomic ranks (100% classified at domain, phylum, and class) and only modest attrition at finer resolution (93.8% classified at genus level).

The phylum-level community composition was dominated by Pseudomonadota (71.6% of classified ASVs), with substantial contributions from Bacteroidota (13.6%) and Campylobacterota (8.0%), and a minor presence of Verrucomicrobiota (2.5%). This distribution is broadly consistent with previously reported bacterial community structures in sea ice and related cryospheric environments, where Pseudomonadota (formerly Proteobacteria) and Bacteroidota are frequently identified as dominant taxa [ref10]. The recovery of these phyla across the experimental sea ice matrix, brine, frost flower, and air/seawater compartments lends support to the interpretation—consistent with the study's stated findings—that niche-based selection processes structure bacterial community assembly during sea ice formation, as evidenced by the emergence of taxa not present in the seeding seawater.

Comparison with Expected Findings

The high rate of taxonomic classification success at broad ranks (100% at domain/phylum/class) aligns with expectations for well-characterized bacterial lineages typically associated with marine and sea ice systems, for which the SILVA database maintains comprehensive reference coverage [ref9]. The modest decline in classification success at the genus level (93.8%) likely reflects the known limitations of short amplicon reads in resolving fine-scale taxonomic distinctions, particularly among less-characterized or environmentally specific lineages that may be underrepresented in reference databases (Jurburg, 2025). This pattern is consistent with general expectations for amplicon-based surveys of environmental bacterial communities rather than indicating any unusual feature of this dataset.

The dominance of Pseudomonadota in the classified ASVs is consistent with prior reports of sea-ice-associated bacterial assemblages, in which taxa within this phylum—including groups associated with psychrophilic and halotolerant lifestyles—are frequently enriched during ice formation (Quast et al., 2013). The secondary abundance of Bacteroidota, a phylum commonly associated with the degradation of complex organic matter and frequently detected in sea ice brine and particulate-associated niches (Eronen-Rasimus et al., 2015), further supports the plausibility of the community structure recovered here. The detection of Campylobacterota, historically associated with microaerophilic or redox-stratified environments [ref5], may reflect the presence of localized geochemical gradients within the ice matrix or brine channels of the experimental chamber, though the amplicon-based approach used here cannot directly confirm the physiological or metabolic activity of these taxa.

Notably, the description of this study emphasizes that bacterial enrichment in sea ice occurred at a rate comparable to prior field-based observations despite minimal diatom representation, and that eukaryotic community composition remained relatively stable across sampling compartments—an outcome interpreted as reflecting stochastic assembly processes for the eukaryotic fraction, in contrast to the niche-based selection apparently governing bacterial distribution. This bacterial/eukaryotic contrast is an important conceptual result, though the taxonomic results presented in this analysis are restricted to the bacterial 16S rRNA gene dataset; the eukaryotic community data referenced in the study description are not detailed in the taxonomic results reported here, and readers should consult accompanying analyses for those findings.

Broader Context

These results contribute to a growing body of literature examining the mechanisms of microbial colonization during sea ice formation, a process with implications for polar biogeochemical cycling, primary production, and the biological pump in ice-covered oceans [ref4]. The use of a controlled laboratory chamber to simulate sea ice formation represents a methodological advance over purely observational field studies, as it allows the decoupling of local ecological and physicochemical drivers (e.g., brine rejection, salinity gradients, temperature) from larger-scale oceanographic and atmospheric influences that typically confound field-based interpretations (Kerfahi et al., 2024). The recovery of archetypal sea-ice-associated bacterial taxa under these controlled conditions, absent from the seeding seawater community, provides experimental support for niche-based selection as a structuring force in this system, complementing correlative evidence from natural sea ice environments [ref2].

Limitations

Several limitations of this study should be acknowledged. First, the overall read retention rate following quality filtering (60.4%) indicates that a substantial fraction of the initial sequencing output was excluded during processing; while this is not unusual for amplicon-based pipelines with stringent quality thresholds, it may have reduced sensitivity to detect rare or low-abundance taxa, particularly in compartments with lower biomass such as frost flowers or air samples. Second, ASV inference and quality filtering were applied to 84 of the 124 total SRA runs, and it is not clear from the available results whether the remaining runs (e.g., additional replicates, alternate marker genes, or non-bacterial datasets) were processed through parallel or complementary pipelines; this should be clarified in future reporting. Third, taxonomic classification was performed using a single reference database (SILVA), and classification accuracy—especially at the genus level—is inherently constrained by the completeness and curation of this database for polar and psychrophilic taxa, which may be underrepresented relative to more commonly studied environments. Fourth, the amplicon-based approach used here provides taxonomic but not direct functional information; while the study describes “functional partitioning” of bacteria into niches, the results presented in this analysis are limited to taxonomic composition, and the functional inferences would require complementary metagenomic or metatranscriptomic evidence not detailed here. Finally, as with all chamber-based experimental systems, some caution is warranted in extrapolating findings to natural sea ice environments, given that laboratory conditions—while designed to mimic key physicochemical gradients—cannot fully replicate the complexity of natural polar ecosystems, including their microbial seed banks, trace nutrient dynamics, and multi-year environmental variability [AUTHOR: additional detail on chamber-to-field extrapolation limitations would strengthen this discussion].

Future Directions

Future work building on these findings could pursue several directions. Shotgun metagenomic or metatranscriptomic sequencing would allow direct assessment of the functional partitioning hypothesized to underlie niche-based bacterial selection, complementing the taxonomic patterns reported here. Extending the temporal resolution of sampling within the chamber system could help clarify the successional dynamics of bacterial colonization during different stages of ice formation and aging. Parallel analysis of the eukaryotic community using amplicon markers appropriate for protistan or algal taxa (e.g., 18S rRNA) would allow more direct testing of the proposed contrast between deterministic bacterial assembly and stochastic eukaryotic assembly. Additionally, comparative analyses using multiple reference databases (e.g., SILVA alongside other curated 16S rRNA databases) could help validate the robustness of genus-level classifications, particularly for taxa of ecological interest in sea ice systems. Finally, replicating this chamber-based experimental approach under varying environmental conditions (e.g., temperature, salinity, or nutrient regimes) would help establish the generalizability of the niche-based selection patterns observed here and clarify the specific environmental variables most responsible for structuring bacterial communities during sea ice formation.

Data Availability

All sequence data are available from the NCBI SRA under accession PRJNA1473294. Analysis code, results, and interactive figures are available in this repository. This paper was generated using the Open Microbial Community platform.

References